

Task-B Report: Multi-Class Deepfake Detection

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1 Problem Statement

The objective of Task B is to develop a robust audio forensic system capable of fine-grained deepfake detection. Unlike binary classification tasks, this challenge requires the model to distinguish between genuine and synthetic audio while simultaneously identifying the gender of the speaker.

Formally, given an input audio waveform $x \in \mathbb{R}^T$, the system must map x to one of four distinct categories $y \in \{0, 1, 2, 3\}$:

1. **Female Real (Class 0):** Female Genuine, unaltered real-world audio recording.
2. **Male Real (Class 1):** Male Genuine, unaltered real-world audio recording.
3. **Female Fake (Class 2):** Synthetic Female audio that has been distorted or degraded.
4. **Male Fake (Class 3):** Synthetic Male audio that has been distorted or degraded.

2 System Architecture

We utilized the **Wav2Vec 2.0** model, a state-of-the-art framework for self-supervised learning of speech representations.

2.1 Backbone Components

- **Feature Encoder:** The model ingests raw waveforms directly. It utilizes a multi-layer convolutional feature encoder to discretize the audio signal into latent speech representations.
- **Context Network:** A Transformer-based network processes the latent representations to capture long-range temporal dependencies.
- **Classification Head:** We appended a linear projection layer on top of the pooled output of the Transformer to map the extracted features to the 4 target classes ($num_labels = 4$).

2.2 Model Initialization

We initialized the system using the pre-trained `facebook/wav2vec2-base` weights. This model has been pre-trained on 960 hours of Librispeech on 16kHz sampled speech audio, providing a robust foundation for feature extraction.

3 Methodology

3.1 Preprocessing Pipeline

To standardize the variable input data, we applied the following preprocessing steps prior to training:

1. **Resampling:** All audio inputs were loaded and re-sampled to 16 kHz to match the pre-training conditions of Wav2Vec 2.0.
2. **Mono Conversion:** Multi-channel audio inputs were averaged to a single mono channel.
3. **Tensor Shaping:** We enforced a fixed input dimension. Audio clips longer than 10 seconds were truncated, while shorter clips were padded with zeros to a length of 160,000 samples ($16\text{kHz} \times 10\text{s}$).

3.2 Training Strategy

We employed a standard fine-tuning approach using the Hugging Face **Trainer** API.

- **Data Splitting:** The training data was split into an 80% training set and a 20% validation set, stratified by the target class to maintain distribution consistency.
- **Objective:** The model was trained to minimize the Cross-Entropy Loss across the four classes.
- **Optimization:** We utilized the AdamW optimizer with weight decay to regularize the model and prevent overfitting.

4 Experimental Setup

The hyperparameters used for fine-tuning the Wav2Vec 2.0 model are detailed in Table 1.

Table 1: Hyperparameter Configuration

Parameter	Value
Input Length	160,000 samples (10s)
Batch Size	8
Optimizer	AdamW
Learning Rate	3×10^{-5}
Weight Decay	0.01
Epochs	2
Precision	FP16 (Mixed Precision)
Evaluation Metric	Accuracy

Implementation was carried out using PyTorch and the Transformers library on a CUDA-enabled GPU.

5 Inference and Results

5.1 Inference Pipeline

For the final submission, we loaded the fine-tuned model from the checkpoint with the highest validation accuracy.

1. **Loading:** The test data was processed using the same pipeline as the training data (16kHz resampling, 10s padding).
2. **Prediction:** We performed batch inference (*batch_size* = 8) on the test set.
3. **Decoding:** The logits were converted to class predictions using the argmax function:

$$\hat{y} = \operatorname{argmax}_{c \in \{0,1,2,3\}} P(y = c|x)$$

6 Conclusion

We successfully adapted the Wav2Vec 2.0 architecture for the multi-class task of deepfake and gender detection. The use of a pre-trained transformer backbone allowed the model to learn discriminative features from raw audio efficiently, even with a limited number of training epochs.